



Cybersecurity and Digital Forensics

At Wilmington University

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Cornerstone Discovery

What is Cybersecurity?

“Cybersecurity is the body of technologies, processes and practices (i.e. measures) designed to protect networks, computers, programs and data from attack, damage or unauthorized access.”

Source: TechTarget

What is a Cyber Threat?

*“A cyber threat is the possibility of a malicious attempt to 1) access files and infiltrate or **steal data** or 2) to **damage or disrupt** a computer network or system.”*

Sources: SecureWorks/Oxford Dictionary

The Statistics Are Alarming

- Hackers attack every 39 Seconds
- 43% of cyber attacks target small business
- Only 14% of small businesses rate their ability to mitigate cyber risks, vulnerabilities and attacks as highly effective
- 60% of small companies go out of business within six months of a cyber attack
- 48% of data security breaches are caused by acts of malicious intent. Human error or system failure account for the rest
- In 2016, 40% of companies expect a data breach caused by malicious insiders

Most Frequent Types of Cyber Threats

- Distributed Denial of Service (DDoS) Attacks
- Insider Threat
- Malware/Spyware
- Password Attacks
- Phishing
- Ransomware

Distributed Denial of Service (DDoS)

- DDoS attacks occur when a server is intentionally overloaded with requests, with the goal of shutting down the target's website or network system



Insider Threat

- Someone with administrative privileges, usually from within the organization, purposely misuses his or her credentials to gain access to confidential company information



Malware/Spyware

- This umbrella term is short for "malicious software," and covers any program introduced into the target's computer with the intent to cause damage or gain unauthorized access



Password Attacks

- 3 main types of password attacks:
 - Brute-force attack, which involves guessing at passwords until the hacker gets in
 - Dictionary attack, which uses a program to try different combinations of dictionary words
 - Keylogging, which tracks all of a user's keystrokes, including login IDs and passwords



Phishing

- Involves collecting sensitive information like login credentials and credit-card information through a legitimate-looking (but ultimately fraudulent) website, often sent to unsuspecting individuals in an email



Ransomware

- Ransomware will either lock you out of your computer and demand money in return for access or threaten to publish private information if you don't pay a specified amount. Ransomware is one of the fastest growing types of security breaches.



Most Common Sources of Cyber Threats

- Nation states or national governments
- Terrorists
- Industrial spies
- Organized crime groups
- Hacktivists and hackers
- Business competitors
- Disgruntled insiders

Source: SecureWorks

Technology Trends Driving Cyber Threats

- Internet of Things
 - Individual devices connected to the internet or other networks (i.e. Amazon Echo, Smart Devices, etc.)
- Explosion of Data (It's everywhere!)
 - Stored in devices such as cellphones, laptops and elsewhere

Main Elements of Cybersecurity

- Ensuring Cybersecurity requires coordinated efforts throughout an information system.
 - Application security
 - Information security
 - Network security
 - Disaster recovery / business continuity planning
 - End-user compliance



Ever Increasing Sources of ESI

Not Just about Desktops, Laptops, Servers, Tablets & Cell Phones



And Now The Internet of Things (IoT)

What is the IoT?

- The IoT is a **Giant Network of Connected "Things"**
- A System of **Interrelated Computing Devices, Mechanical and Digital Machines, Objects, Animals or People** That are Provided With Unique Identifiers and the Ability to Transfer Data Over a Network Without Requiring Human-to-Human or Human-to-Computer Interaction



Pertinent Questions

- Where Was Someone at any Point in Time?
- How Fast Was Someone Travelling and What Route Did They Take?
- When Was Someone Using One of Their Electronic Devices?
- What, in Particular, Were They Doing On Their Device at any Point in Time?
- With Whom Were They Communicating On Their Device at any Point in Time?

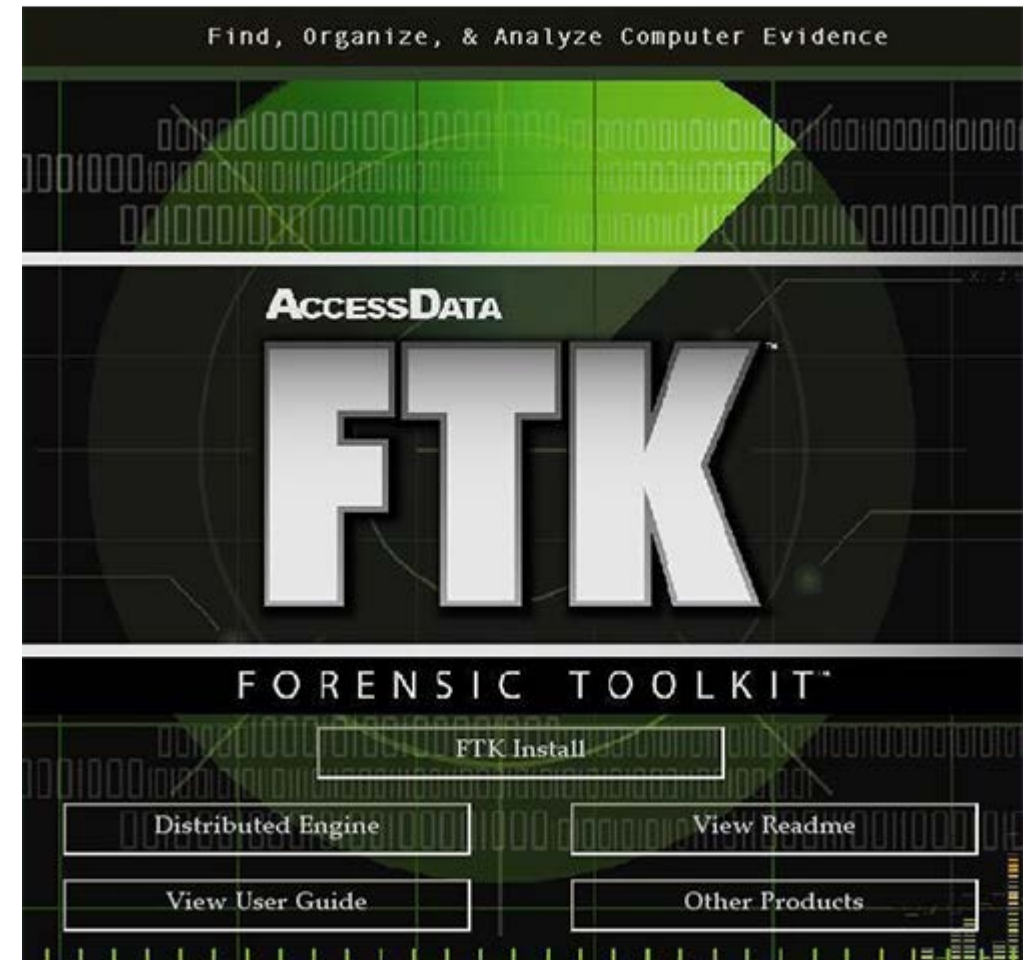
Digital Forensics!

cellebrite
delivering mobile expertise

nuix
Simple. Powerful. Precise.

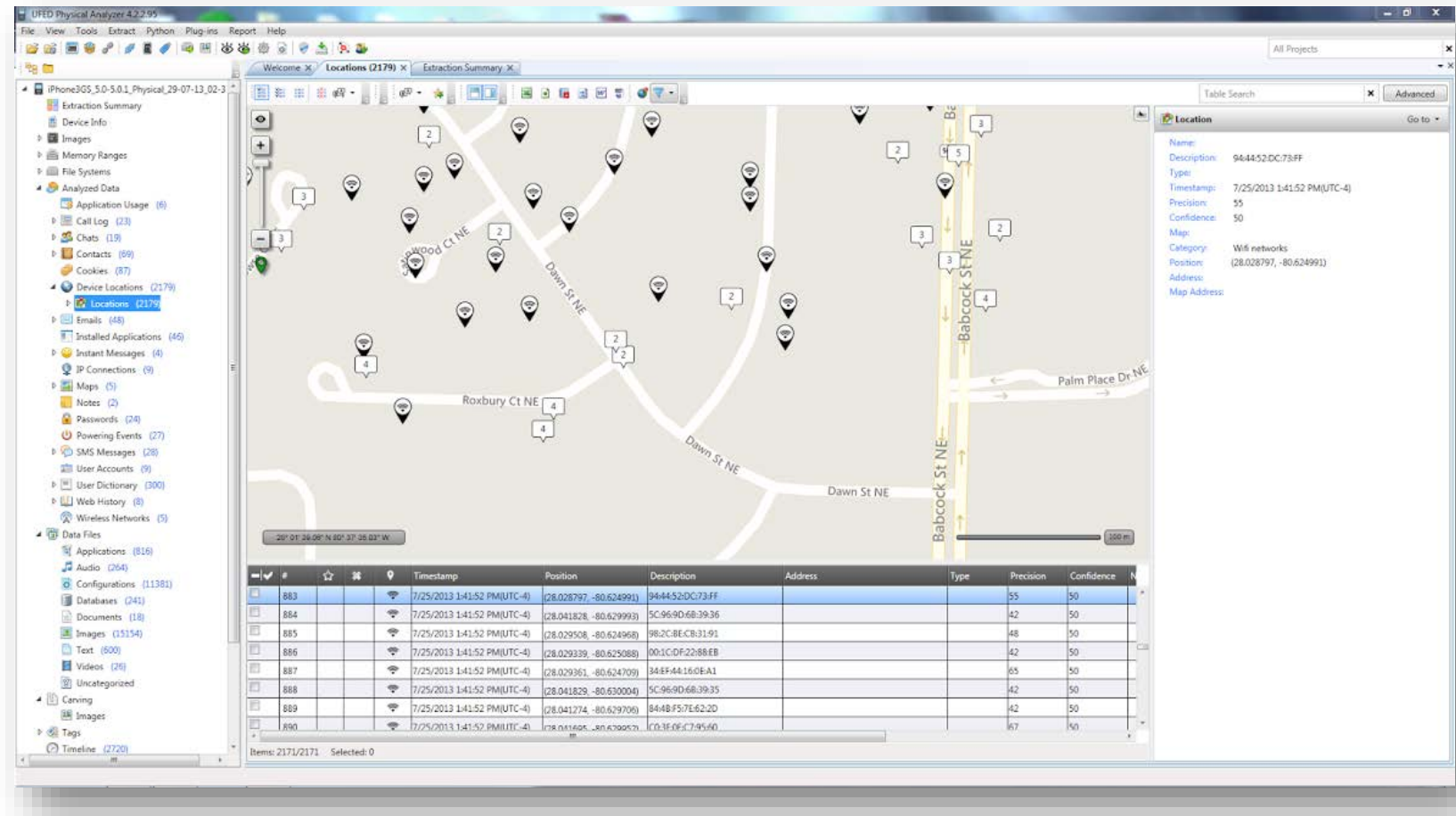
MAGNET AXIOM™

FOR COMPUTERS



Where Was Someone at any Point in Time?

- Remember ...Not Just Laptops, Cell Phones, Tablets, and GPS
- **IoT Devices Communicate Over Cell Networks too**



Where Was Someone at any Point in Time?

- Pictures and Videos – EXIF Data – Date, Time, Location ..and More!
- Free EXIF Viewer at: thetechgazette.com

☒ From Web ☐ From File Image URL:

Basic Image Information

Target file: 20130608_220756.jpg

Camera:	Samsung SCH-I545
Lens:	4.2 mm (Max aperture f2.2) (shot wide open)
Exposure:	Auto exposure, Program AE, 1/17 sec, f/2.2, ISO 640
Flash:	none
User Comment:	METADATA-START
Date:	June 8, 2013 10:07:56PM (timezone not specified) (2 years, 3 months, 28 days, 8 hours, 43 minutes, 11 seconds ago, assuming image timezone of US Pacific)
Location:	Latitude/longitude: 39° 56' 59.7" North, 75° 10' 7.7" West (39.949913, -75.168816) Location guessed from coordinates: <i>1629-1631 Walnut St, Philadelphia, PA 19103, USA</i> Map via embedded coordinates at: Google , Yahoo , WikiMapia , OpenStreetMap , Bing (also see the Google Maps pane below) Altitude: 4,294,874 meters (14,090,795 feet) below sea level
File:	4,128 × 2,322 JPEG (9.6 megapixels) 3,885,629 bytes (3.7 megabytes)
Color Encoding:	WARNING: Color space tagged as sRGB, without an embedded color profile. Windows and Mac browsers and apps treat the colors randomly.

[CLEAR IMAGE]

Extracted 512 × 288 33-kilobyte "Composite:ThumbnailImage" JPG
Displayed here at 88% width ($\frac{1}{84}$ the area of the original)



Click image to isolate; click this text to show histogram

Main JPG image displayed here at 11% width ($\frac{1}{84}$ the area of the original)



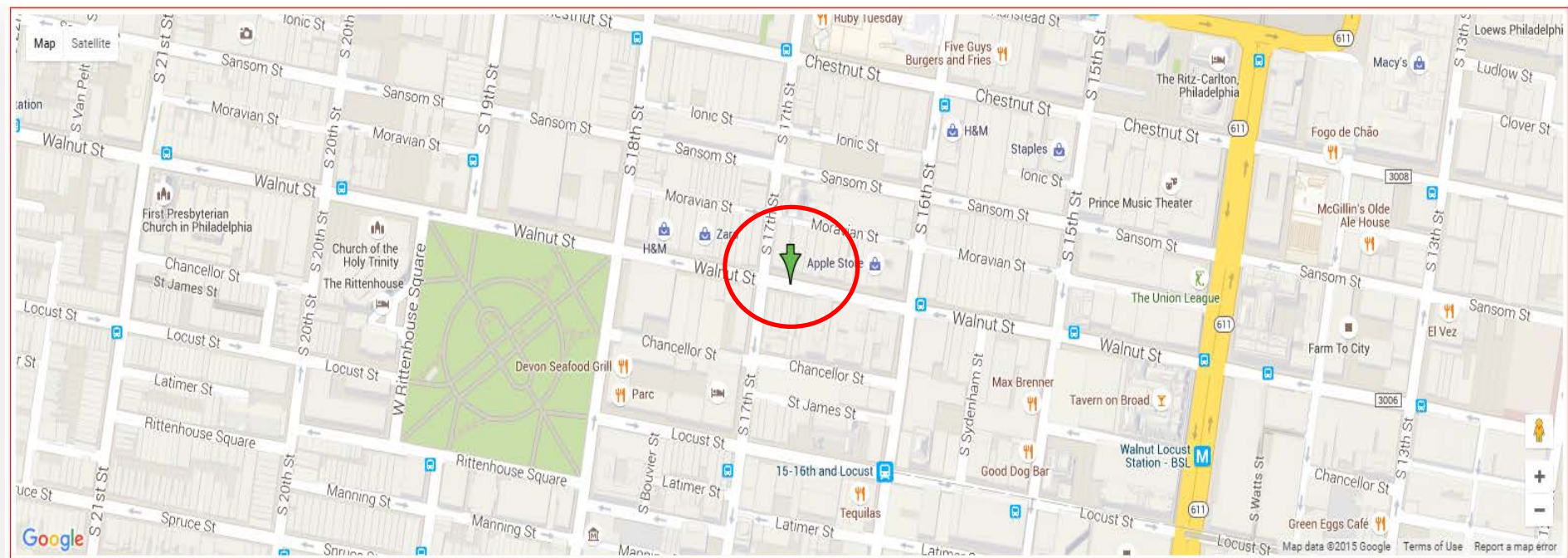
Where Was Someone at any Point in Time?

- EXIF Data



Where Was Someone at any Point in Time?

- Location information



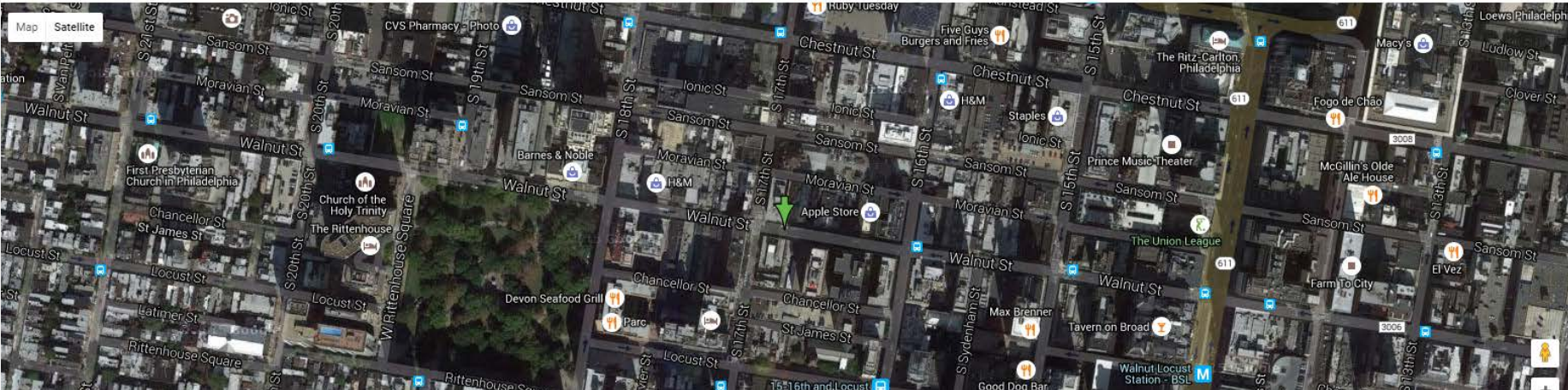
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Color Encoding:	WARNING: Color space tagged as sRGB, without an embedded color profile. Windows and Mac browsers and apps treat the colors randomly. Images for the web are most widely viewable when in the sRGB color space and with an embedded color profile. See my Introduction to Digital-Image Color Spaces for more information.

Main JPG image displayed here at 11% width ($1/8$ the area of the original)



Click image to isolate; click this text to show histogram



How Fast was Someone Travelling?

What Route did They Take?

- Google Location History

Location history

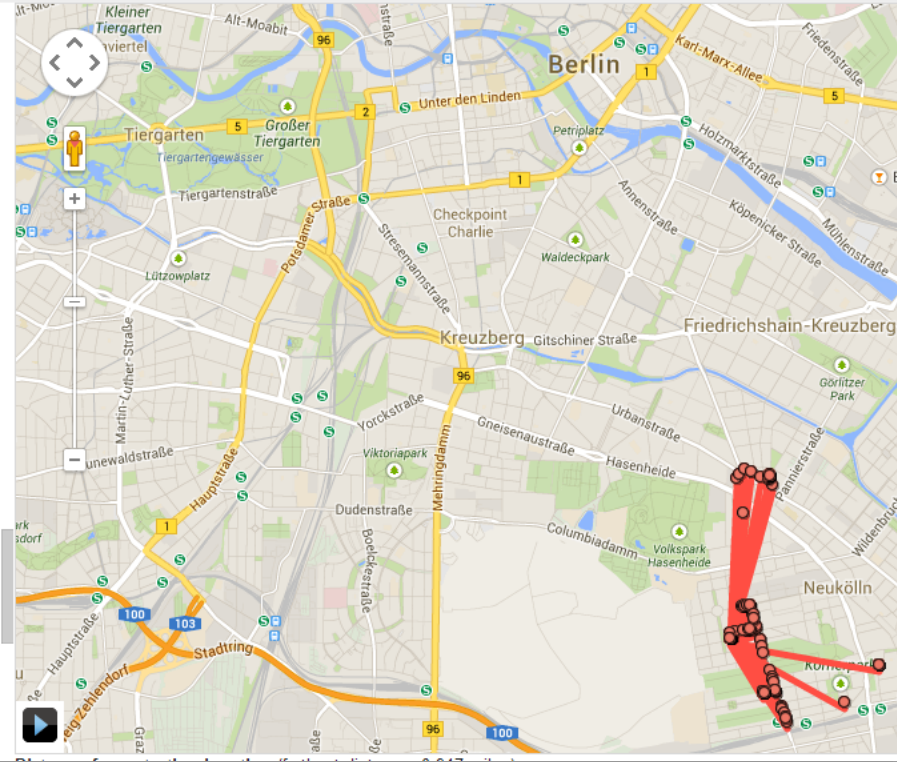
January 2014						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Show: 1 Day ▾

7:00 PM - 8:00 PM
7:13 PM

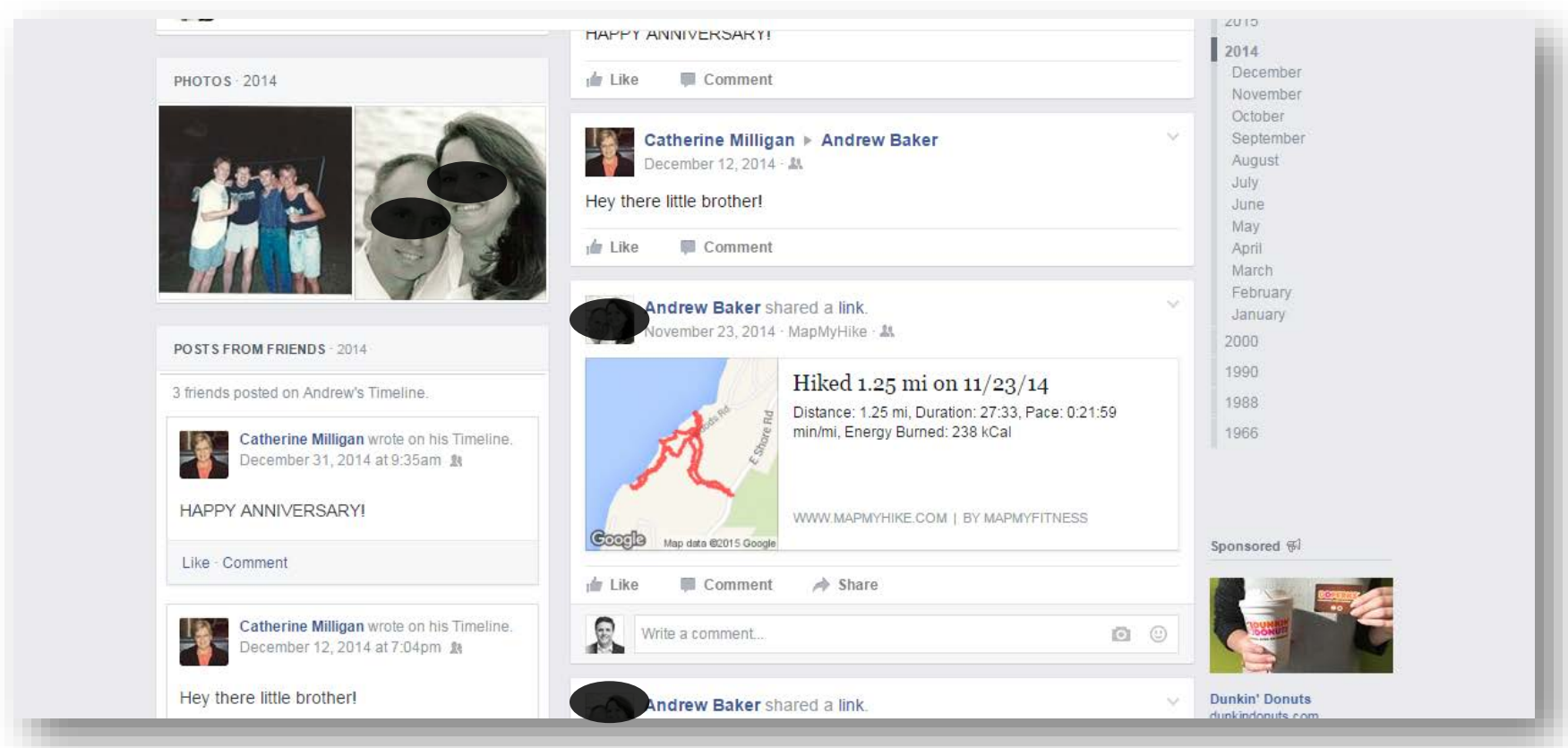
8:00 PM - 9:00 PM
8:06 PM 8:33 PM

9:00 PM - 10:00 PM
9:06 PM 9:07:23 PM 9:07:27 PM
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9:10:34 PM 9:10:40 PM 9:11 PM
9:12:01 PM 9:12:37 PM 9:13:01 PM
9:13:41 PM 9:14:02 PM 9:14:44 PM
9:15:02 PM 9:15:43 PM 9:16:33 PM
9:16:43 PM 9:18 PM 9:19 PM 9:21 PM
9:23:11 PM 9:23:21 PM 9:24 PM
9:25:21 PM 9:25:42 PM 9:26:21 PM
9:26:43 PM 9:27 PM 9:28:05 PM
9:28:24 PM 9:29:05 PM 9:29:45 PM
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9:32:25 PM 9:34 PM 9:38:02 PM
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9:40:04 PM 9:40:50 PM 9:41:04 PM
9:41:51 PM 9:42 PM 9:44:01 PM
9:44:05 PM 9:46:03 PM 9:46:21 PM
9:47:08 PM 9:47:54 PM 9:48:07 PM
9:48:54 PM 9:49:08 PM 9:49:56 PM



How Fast Was Someone Travelling? What Route did They Take?

- Social Media



How Fast was Someone Travelling?

What Route did They Take?

- Call Detail Records - CDR

Combined Detail LE [REDACTED]-384-1700

MSISDN	Start Time	First Tower Cell ID / ECGI Cell	First Tower Latitude	First Tower Longitude	Last Tower Cell ID / ECGI Cell	Last Tower Latitude	Last Tower Longitude	Duration(Seconds)
43841700	8/23/2014 02:13:55 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	43
43841700	8/23/2014 02:15:40 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	48
43841700	8/23/2014 02:17:15 AM	15523	42.11114849	-80.0794776	16012	42.11005556	-80.1105556	23
43841700	8/23/2014 02:28:24 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	2
43841700	8/23/2014 02:28:47 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	99
43841700	8/23/2014 02:31:54 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	41
43841700	8/23/2014 02:32:35 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	28
43841700	8/23/2014 02:33:38 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	34
43841700	8/23/2014 02:34:13 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	45
43841700	8/23/2014 02:35:54 AM	15513	42.12546944	-80.0793222	15523	42.11114849	-80.0794776	37
43841700	8/23/2014 02:43:33 AM	16012	42.11005556	-80.1105556	16012	42.11005556	-80.1105556	82
43841700	8/23/2014 02:48:27 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	15
43841700	8/23/2014 02:52:44 AM	16012	42.11005556	-80.1105556	15523	42.11114849	-80.0794776	37
43841700	8/23/2014 02:53:24 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	22
43841700	8/23/2014 02:54:42 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	34
43841700	8/23/2014 02:55:18 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	34
43841700	8/23/2014 02:56:21 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	32
43841700	8/23/2014 02:57:09 AM	15523	42.11114849	-80.0794776	15523	42.11114849	-80.0794776	35
43841700	8/23/2014 02:58:06 AM	15513	42.12546944	-80.0793222	15513	42.12546944	-80.0793222	22

How Fast was Someone Travelling? What Route did They Take?

- IoT Devices That can Communicate Over Cell Networks



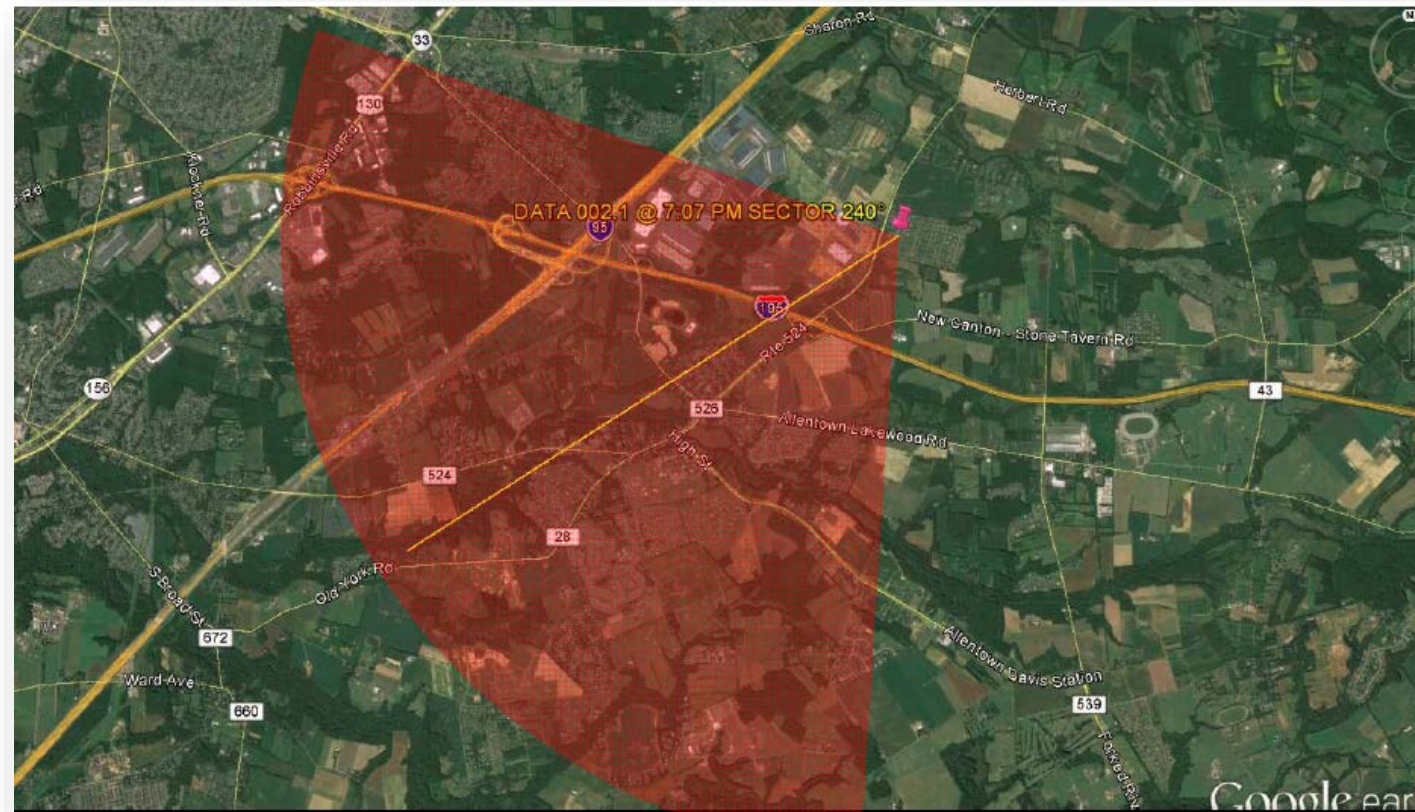
T-Mobile®



ZTE Quartz™

How Fast was Someone Travelling? What Route did They Take?

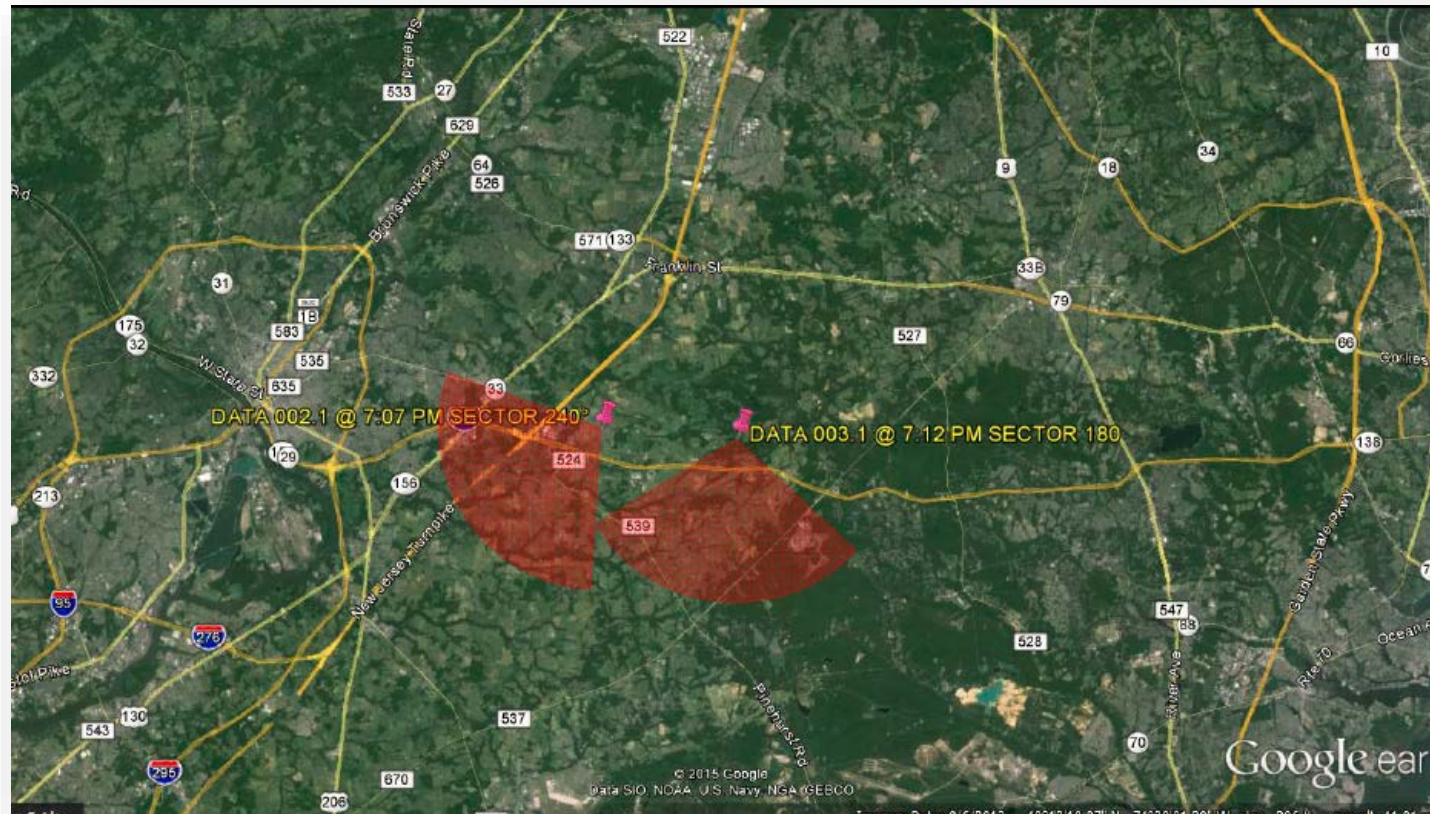
- Call Detail Records - CDR



How Fast was Someone Travelling?

What Route did They Take?

- Call Detail Records - CDR



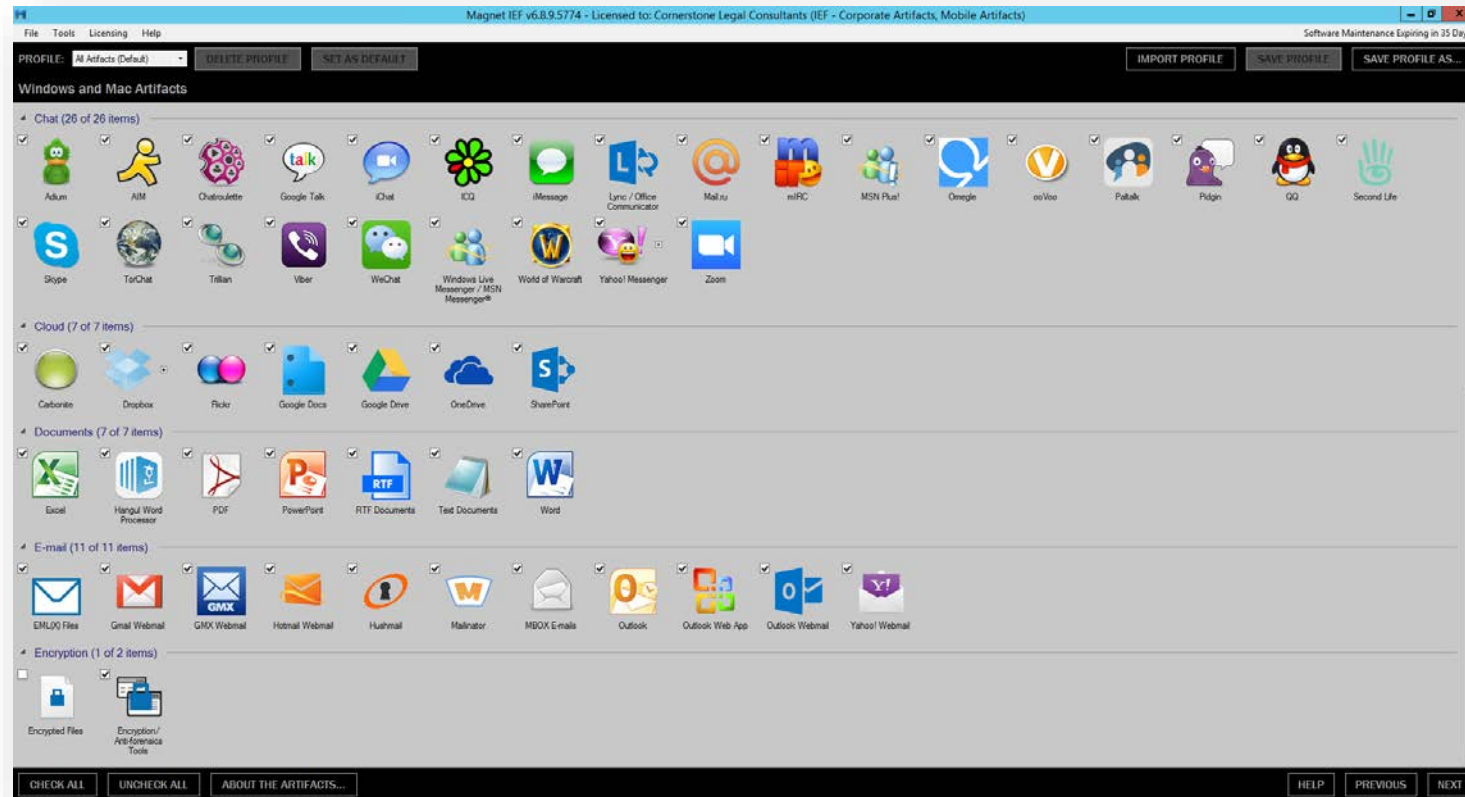
When was Someone Using one of Their Electronic Devices?

- Timeline of Device Activity Obtained via Cell Phone

#	Title	URL	Last Visited	Visits	Source	Deleted
1		https://m.facebook.com/login.php?refsrc=facebook.com%2Fmessages%2F&lwv=100&refid=8	10/12/2015 6:03:05 PM(UTC-4)	1	Safari	
2	Facebook	https://m.facebook.com/home.php?refsrc=facebook.com%2Fmessages%2F&refid=8&_rdr	10/12/2015 6:03:05 PM(UTC-4)	1	Safari	
3	Giovanni [REDACTED]	https://m.facebook.com/messages/read/[REDACTED]6%3A80c8af0dd5aa004790	10/12/2015 6:03:16 PM(UTC-4)	1	Safari	
4	[REDACTED] McDonnell	https://m.facebook.com/messages/read/[REDACTED]2ca16416dd531beef32340	10/12/2015 6:03:26 PM(UTC-4)	1	Safari	
5	Facebook	https://m.facebook.com/home.php?refsrc=facebook.com%2Fmessages%2F	10/12/2015 6:04:02 PM(UTC-4)	5	Safari	
6	Facebook	https://m.facebook.com/home.php?refsrc=facebook.com%2Fmessages%2F&soft=messages	10/12/2015 6:04:03 PM(UTC-4)	3	Safari	
7	Monica [REDACTED]	https://m.facebook.com/messages/read/[REDACTED]90	10/12/2015 6:04:07 PM(UTC-4)	1	Safari	
8	Monica [REDACTED]	https://m.facebook.com/messages/read/[REDACTED]90&soft=bookmarks	10/12/2015 6:04:49 PM(UTC-4)	1	Safari	
9		https://m.facebook.com/logout.php?h=Aff[REDACTED]4687386&refid=7	10/12/2015 6:04:54 PM(UTC-4)	1	Safari	

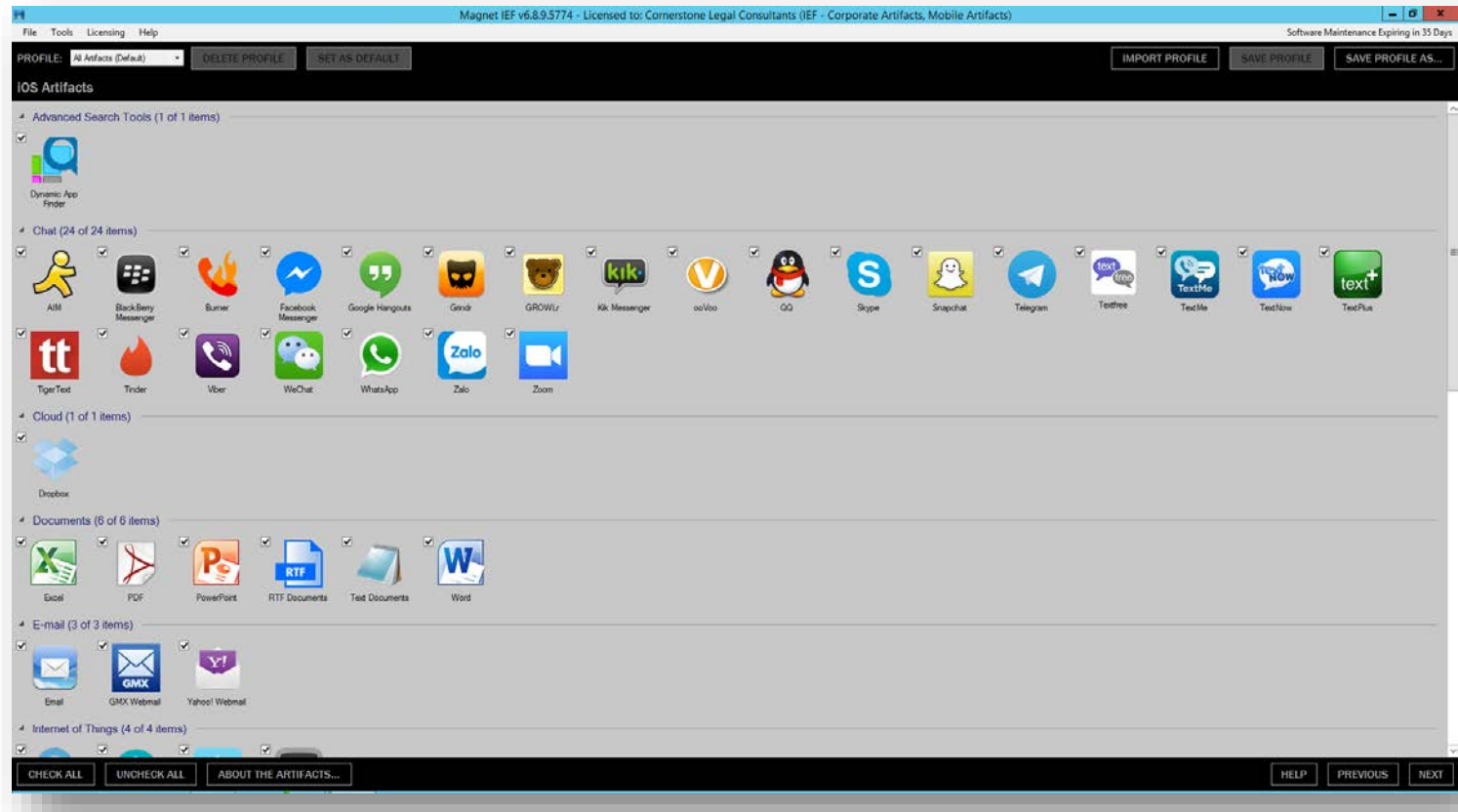
What, in Particular, Were They Doing on Their Device in any Point in Time?

- Internet History



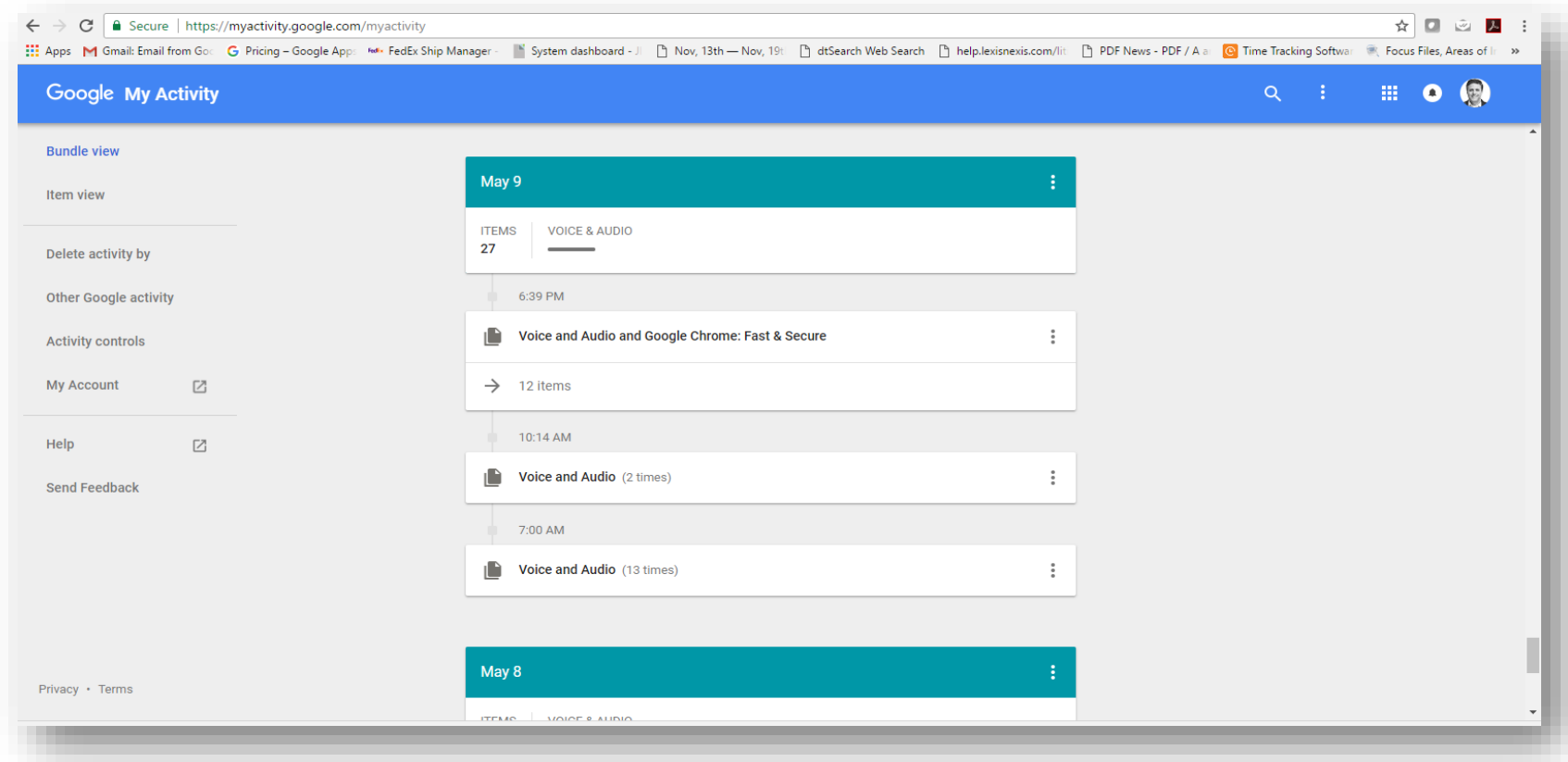
What, in Particular, Were They Doing on Their Device in any Point in Time?

- Internet History



What, in Particular, Were They Doing on Their Device in any Point in Time?

- Google “My Activity”



What, in Particular, Were They Doing on Their Device in any Point in Time?

- Google “Takeout”

Select data to include

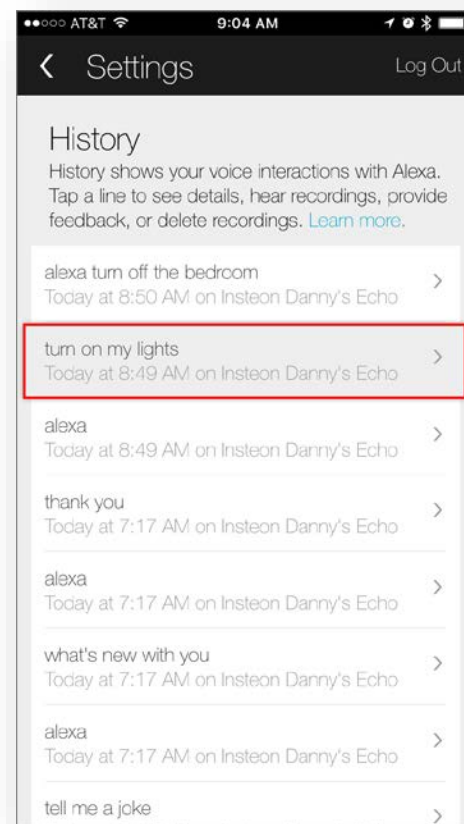
Choose the Google products to include in your archive and configure the settings for each product. This archive will only be accessible to you. [Learn more](#)

Product	Details	Select none
+1s		☒
Bookmarks		☒
Calendar	All calendars	☒
Chrome	All Chrome data types	☒
Contacts	vCard format	☒
Drive	All files Microsoft Powerpoint and 3 other formats	☒
Fit	All data types	☒
Google Photos	All photo albums	☒
Google Play Books	All books HTML format	☒
Google+ Circles	vCard format	☒
Google+ Pages	All pages HTML format	☒
Google+ Stream	HTML format	☒
Groups		☒

Handsfree		☒
Hangouts		☒
Hangouts on Air		☒
Keep		☒
Location History	JSON format	☒
Mail	All mail	☒
Maps (your places)		☒
Moderator		☒
My Maps		☒
Profile		☒
Searches		☒
Tasks		☒
Voice		☒
Wallet		☒
YouTube	All data types OPML (RSS) format	☒

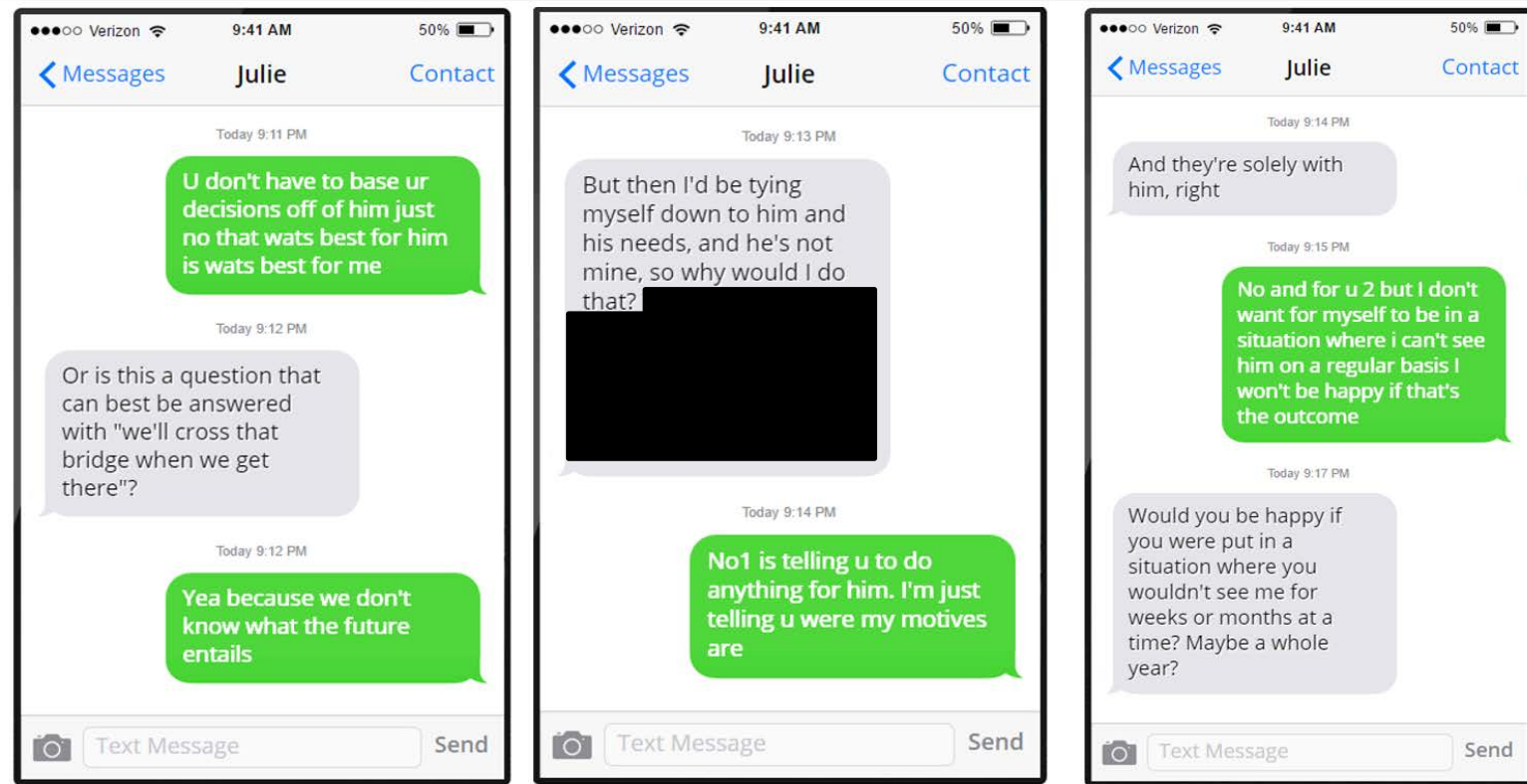
What, in Particular, Were They Doing on Their Device in any Point in Time?

- Activity Reported via Amazon Echo & Dot – Alexa



With Whom Were They Communicating on Their Device at any Point of Time?

- Text Activity Obtained via Cell Phone



Thank you!



This Presentation and Other Information Available at:

thetechgazette.com